

MNEMO

The second brain that lives on your machine —
and lets your agents in, on your terms.

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LOCAL-FIRST · OFFLINE FOREVER · YOURS

1 What Mnemo is

Mnemo is a **second brain**: a personal knowledge vault of notes that lives entirely on your own computer, in a folder of plain Markdown files you can open with anything. It is named for **Mnemosyne**, the Greek goddess of Memory and mother of the Muses. Its emblem — a coil rising out of her spring, like a strand of DNA — is not decoration: **the mark and the graph are the same object**. Your notes appear as strands rising through time, linked by the ideas they share.

Mnemo was built for the machines the big tools keep abandoning. It runs on hardware and operating systems that current second-brain apps no longer support — its reference floor is a 2012 laptop on macOS 10.15 — with no account, no telemetry, and no cloud standing between you and your first note. It works completely offline, forever, and it costs nothing to run.

And it has one capability no other second brain ships natively: **your AI agents are citizens of the vault**. Through the Model Context Protocol (MCP), an agent can search, read, and learn your notes — but every change it wants to make arrives as a **proposal** that only you can approve. Mnemo is a brain built for a human and their crew, with the human holding the gate.

Files are truth

Your vault is an ordinary folder of `.md` files. Mnemo's index is a rebuildable cache — delete it and nothing is lost. Any editor, file manager, or backup tool can touch the same files. No lock-in, ever.

The Gate

Agents never write directly. An agent's change is stored as an exact, reviewable proposal; nothing lands in your vault until you approve it. There is no bypass — by design, not by setting.

Keys, not accounts

Access is granted with keys you create and can revoke: **person keys** for humans, **agent keys** for software. There is no sign-up, no server, and nothing to phone home to.

The Helix

Your notes arranged as strands on a time axis, paired with their strongest links and traced with similarity arcs. Gold appears in exactly one place: where an agent's proposal is waiting for you.

2 Getting started

What you need

- **Node.js 20 LTS** — Mnemo's only runtime. (Newer Node lines drop old operating systems; Node 20 is the line that still honours the floor.)
- A machine Mnemo is built to respect — the reference floor is **macOS 10.15.7** on a 2012 MacBook Pro; anything newer is comfortable.
- A folder for your vault. New and empty is fine; a folder of existing Markdown becomes a vault the moment Mnemo points at it.

Install

```
git clone https://github.com/ronin1102-tech/Mnemo.git
cd Mnemo
npm run install:all
npm run build
```

First run

On the Mac, open **Mnemo.app** — it starts Mnemo in the background if it isn't running and opens the Helix, no terminal to babysit. From the command line, the equivalent is:

```
node packages/mnemo-mcp/dist/ctl.js --vault ~/Mnemo-Vault ui
```

The `ui` command prints a local link holding your session key. Open it, and write your first note from the **Read** tab (see §4). That note is now a file in your vault — open the folder and look: Mnemo's first promise, kept in plain sight.

> **The vault is just a folder.** Notes are `.md` files; folders are folders. Mnemo keeps its own housekeeping (index, logs, proposal store) in a `.mnemo` subfolder inside the vault. Back up the folder and you have backed up everything.

3 The four views

The Helix interface has four tabs. Everything you do in Mnemo happens in one of them.

View	What it's for
Emblem	The mark, large: your vault as the spring and the rising coil. The front door of the app — and a quiet status surface: if gold appears here, something is waiting for your decision.
Helix	Your notes as strands on a vertical time axis. The thinking view — see §3.1.
Graph	A classic force-directed map of notes and links, for when you want the familiar overview.
Read	Where notes are read and written: the strand list, the reader, and the owner editor — **+ Note**, **+ Request**, and **Edit**. Proposals awaiting you are reviewed and settled here and in the Helix.

3.1 Reading the helix

- **Time flows upward.** The oldest notes rest in the spring at the bottom; the newest grow at the top. A note's date is its frontmatter `created` date when it has one, and its file date otherwise.
- **Two strands, paired by strength.** Each note sits on a strand beside its strongest link — the note it shares the most neighbours with, most recent first — so related thinking literally twines together.
- **Rungs are links.** The bars crossing between strands are `[[wikilinks]]` between notes. Fainter arcs are similarity: notes the index judges kin by shared links and tags, always explainable from the underlying data.
- **Focus mode.** Select a note and its links light up while the rest dims — the neighbourhood of one thought, on demand.
- **Gold means one thing.** A gold marker is an agent's pending proposal. Nothing else in the interface uses gold (the emblem's own gold is part of the mark). See gold, and there's a decision waiting.

> **The strand list** under the helix is the same notes in plain list form — newest first, with dates — for scanning and for smaller screens.

4 Writing as the owner

The gate governs agents. **It does not govern you.** When you write in Mnemo, your words land directly in the vault — no proposal, no approval step — because the vault is yours.

The editor

- **+ Note** (Read tab) creates a note: give it a title and text, and choose Create note. The note exists as a file the moment you do.
- **Edit** opens any note in the editor. A preview sits beside your text so you can see the rendered note as you write.
- Press `⌘S` to save an edit. Saves are **byte-for-byte and atomic**: what you typed is what lands on disk — tabs, unicode, line endings — written in one all-or-nothing motion, so a crash can never leave half a note.
- **Conflicts are a conversation, not a casualty.** If the file changed on disk while you were editing (another editor, an approved proposal), Mnemo stops and shows you the choice rather than overwriting. Nothing is silently lost.
- Leave a note with unsaved changes and Mnemo asks before discarding them. Pressing `Esc` backs out of the editor safely.

Notes are limited to **10 MB** each — generous for text, and the limit fails cleanly with an explanation, never a silent truncation.

Writing from your phone

With the hub running (§8), the same editor works from your phone's browser over your private network: sign in with your person key and the Read tab offers the full editor — **+ Note**, **+ Request**, **Edit**. Every write you make is recorded in the audit log under your key's name.

5 Requests — tasking your crew

A **request** is how you ask the agents who work in your vault to do something: summarise a note, draft an outline, research a thread. You write requests inside Mnemo, with **+ Request** in the Read tab. A request is itself a note, kept in the vault's `Requests/` folder — your inbox to the crew, stored as data you own.

```
you write a request  →  an agent reads it  →  the agent answers with a proposal  →  you approve  →  the work lands
```

- **open** — written, waiting for an agent to pick it up.
- **in-progress** — an agent has answered with a proposal that is waiting for your decision.
- **addressed** — you approved the answer; the work has landed.
- **closed** — you marked the request done (edit it and set `status: done` in its frontmatter).

> **Requests are agent-proof.** No agent can write, edit, or delete anything in `Requests/` — agents may only answer a request, by proposing work elsewhere in the vault. Your instructions to the crew can never be quietly rewritten by the crew.

6 The Gate — how agent writes work

Every change an agent makes to your vault passes through the same four beats:

```
propose → pending → you review the diff → approve (lands byte-for-byte) or reject
```

- **Exact bytes.** A proposal stores the precise content proposed. What you approve is what lands — character for character, verified.
- **Stale halt.** If the note changed after the proposal was made, the proposal goes stale and cannot land. The agent must re-propose against the current note. Your newer words are never overwritten by an older suggestion.
- **Expiry.** Unsettled proposals expire after **7 days**. The gold marker on the helix is the countdown you can see.
- **Attribution.** Every settlement records who settled it (`settledBy`) in the vault's audit log — a complete, inspectable history of every write, by you and by agents, lives in `.mnemo/logs/` .
- **Delete means trash.** An approved deletion moves the note to the trash; permanent destruction is a separate, deliberate command (`mnemo-ctl purge-trash`) that only you can run.

You settle proposals where you see them — in the Helix interface, from your computer or your phone, or with `mnemo-ctl approve` . There is no approve tool on the agent side of the wall, and no flag, anywhere, that skips the gate.

7 Keys — who can do what

Mnemo has no accounts. Access is a **key**: a named secret you create, hand to a person or a program, and can revoke at any moment. Only a key's hash is stored in your vault — the value itself is shown once, when created.

Who	Read	Settle proposals	Write notes
Local session (Mnemo.app / this computer)	✓	✓	✓ — directly
Person key (you, on your phone)	✓	✓	✓ — directly, logged under the key's name
Read-only person key	✓	✓	—
Agent key (MCP clients)	✓	—	Proposals only — the Gate decides

```
# create keys (the value is shown once — keep it like a password)
mnemo-ctl token add phone --person
mnemo-ctl token add research-agent --agent
mnemo-ctl token add guest --person --read-only

mnemo-ctl token list        # who holds keys
mnemo-ctl token revoke phone # ends that key on its next request
```

> **Granting a key is granting the hand.** A person key writes as you. Create one per device or per person, so a lost phone is a revocation, not a disaster. Agent keys are the safe thing to hand out freely — the worst an agent can do is ask.

8 Your phone & other machines — the hub

One machine holds the vault and runs the **hub**. Every other device joins it over **Tailscale**, the private network that links your own machines — no public exposure, no port forwarding, no cloud relay.

```
# on the vault machine: serve the Helix to your tailnet
mnemo-ctl --vault ~/Mnemo-Vault hub --bind <mac-tailnet-ip> \
    --allow-host <mac-magicdns-name>

# optional: start it automatically at login, restart if it stops
mnemo-ctl --vault ~/Mnemo-Vault launcher --hub --bind <mac-tailnet-ip> \
    --allow-host <mac-magicdns-name> --at-login
```

- **Your link** looks like `http://<mac-magicdns-name>:7337/#token=<your person key>`. Open it on your phone and the header shows signed in as <your name>. Bookmark it; the second brain now lives in your hand.
- The hub answers **only** to your tailnet address and its MagicDNS name — any other Host is refused (403). Port 7337 is the default; pick another with `--port` if it's taken.

- Opening **Mnemo.app** on the vault machine also starts the hub in the background if it isn't running.
- Agents reach the same machine over MCP (§9) — the hub is the people's door; MCP is the agents' door; the vault behind both is one and the same.

9 Agents & MCP

Mnemo speaks the **Model Context Protocol** natively — the door is in the box, not bolted on by a third-party plugin. Point any MCP client at the same server command the app itself uses:

```
node /path/to/Mnemo/packages/mnemo-mcp/dist/cli.js --vault /path/to/vault serve
```

With an agent key, the client gets sixteen tools. Reading is free; **every writing tool creates a proposal** (§6) — an agent using `write_note` has not written anything. It has asked.

Tool	What it does
`search_notes`	Full-text search across the vault (FTS5).
`read_note`	Read one note, with its links and backlinks.
`list_notes` · `list_recent`	Browse the vault; newest first.
`get_backlinks` · `get_neighbors`	What links here; the neighbourhood of a note.
`find_similar`	Notes the index judges kin to a given note.
`get_tags`	The vault's tag landscape.
`daily_note`	Look up today's dated note (read-only lookup).
`list_requests`	Your open requests to the crew, with their status.
`list_proposals`	Pending and settled proposals.
`write_note` · `append_note` · `patch_note`	Propose a new note, an addition, or a targeted edit.
`move_note`	Propose a move/rename — links across the vault are rewritten to follow.
`delete_note`	Propose a deletion (lands in the trash, never oblivion).

> **Answering a request:** an agent answers by proposing its work with the request named in the proposal — which flips your request to in-progress until you settle it. The request file itself is never modified by the system.

10 mnemo-ctl & daily recipes

The command surface, at a glance

Command	What it does
`mnemo-ctl ui`	Serve the Helix locally (loopback session).
`mnemo-ctl hub`	Serve the Helix to your tailnet (§8).
`mnemo-ctl launcher --at-login`	Start the hub at login; restart it if it stops.
`mnemo-ctl stop`	Stop the background Mnemo.
`mnemo-ctl token …`	Add, list, and revoke keys (§7).
`mnemo-ctl list / approve / reject`	Inspect and settle proposals from the terminal.
`mnemo-ctl show`	Inspect a request and its answers.
`mnemo-ctl export-graph`	Write a static snapshot of the graph (read-only by nature).
`mnemo-ctl purge-trash`	Permanently destroy trashed notes. Deliberate, manual, yours.

Recipes

- **Morning note.** Open the daily note, dump the day into it. Agents with `daily_note` can find it without being told where it is.
- **Capture from anywhere.** Phone → hub → **+ Note**. If it's a job for the crew instead, **+ Request** and let an agent draft it for your approval.
- **Evening review.** Look for gold. Settle the day's proposals while the diffs are small — the gate is a habit, not a chore, when it's daily.
- **Rename without fear.** Moves rewrite wikilinks and Markdown links across the vault to follow the note — rename the file, keep the web.

11 When something's off

Symptom	What's true, and what to do
The hub says it doesn't have that address	Tailscale isn't connected, or its address changed. Reconnect Tailscale, check the machine's tailnet IP, start the hub again.
403 from the hub	You're reaching it by a name or address it wasn't told to allow. Use the MagicDNS name or the tailnet IP you bound, or add the name with `--allow-host`.
"Missing or wrong key"	The link needs your person key after `#token=`, or the key was revoked. Check `token list`; mint a fresh key if the old one is lost — values are shown only once.
No editor on the phone	You're browsing without a person key, or with a read-only key. Open the signed-in link (§8) with a write-capable person key.
A proposal won't land	It's stale (the note changed since) or expired (>7 days). That's the gate working: ask the agent to re-propose against the current note.
An agent wrote to my vault?!	It didn't — it proposed. Nothing lands without your approval; check `.mnemo/logs/` to see exactly what was settled, by whom.

12 Where Mnemo is going

Everything above this page is Mnemo today. This page is the map forward — **roadmap, not current features** — sequenced in waves, each shipped only when it passes on the reference hardware:

- **Wave A — Foundations:** typed properties with a form editor, aliases, the search operator grammar (`tag:`, `path:`, property queries), embeds and transclusion, callouts, footnotes, visual table editing, attachments as citizens.
- **Wave B — Daily-driver parity:** quick switcher, command palette, slash commands, page preview, bookmarks, unlinked mentions, templates, daily-notes flow, snapshots and file recovery, outline panel, local graph.
- **Wave C — Stand-out surfaces:** database-style views over your properties, an infinite canvas, importers (Notion, Evernote, Apple Notes) and a web clipper, Windows and Linux builds on the same old-hardware floor, and static export — publish your own notes, on your own hosting.
- **Wave D — Ecosystem:** themes and CSS snippets, a plugin API, computed properties and formulas.

The rules that never bend

- Files are truth; the index is a cache.
- Every agent write is a human-approved proposal. No bypass exists.
- No accounts, no telemetry, no ads — by architecture.
- The local vault is never crippled to sell you the cloud. If a paid cloud ever exists, it will be optional, and the local brain will stay whole.
- The floor is the mission: Mnemo is built first for the thinkers and ponderers on the machines everyone else abandoned.

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